

PURE 2. Functions

* numbers review

Natural numbers $\mathbb{N}$ (e.g.: 1, 2, 3)

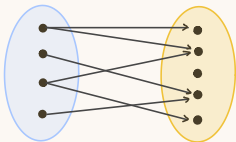
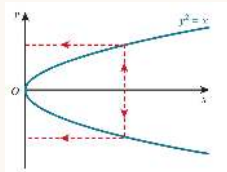
Integers $\mathbb{Z}$ (e.g.: -3, -2, -1, 0, 1, 2, 3)

Rational numbers $\mathbb{Q}$ (e.g.: $\frac{2}{3}$, $\frac{11}{5}$)
= anything displayed as fractions

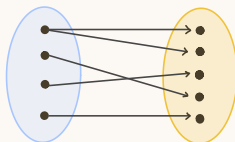
Real numbers $\mathbb{R}$ = all rational + irrational #s (e.g.: $\sqrt{2}$, π)

1. Mapping fundamentals

= input $\rightarrow$ output
object $\rightarrow$ image
domain $\rightarrow$ range



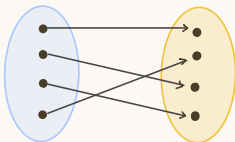
many - to - many



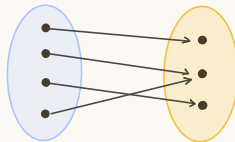
one - to - many

A function is a

one - to - one mapping

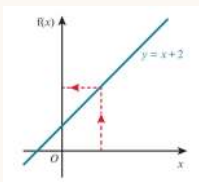


many - to - one mapping



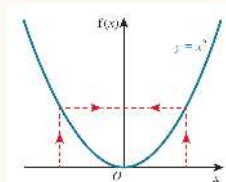
$$f: x \mapsto x+2 ; x \in \mathbb{R}$$

OK $f(x) = x+2 ; x \in \mathbb{R}$



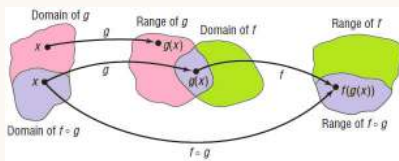
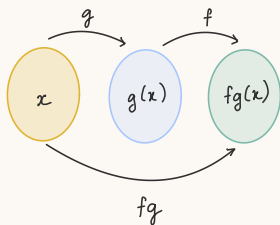
$$f: x \mapsto x^2 ; x \in \mathbb{R}$$

OK $f(x) = x^2 ; x \in \mathbb{R}$



2 Composite functions

= combination of ≥ 2 functions



- in general, $f(g)x \neq gf(x)$
- $ff(x)$ = function applied twice

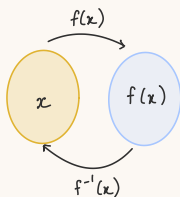
e.g.: $f(x) = 2x + 3$; $g(x) = x^3$; Find $gf(2)$

$$x \xrightarrow{f} 2x + 3 \xrightarrow{g} (2x + 3)^3$$

$$2 \xrightarrow{f} 7 \xrightarrow{g} 343$$

3. Inverse functions

= undoes what the function does



- $ff^{-1}(x) = f^{-1}f(x) = x$
- the domain of $f^{-1}(x)$ is the range of $f(x)$
- the range of $f^{-1}(x)$ is the domain of $f(x)$
- * not every function has an inverse
 - ↳ only if $f(x)$ is a one-one mapping does $f^{-1}(x)$ exist

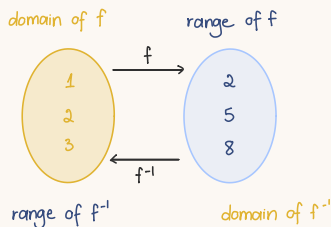
e.g.: $f(x) = 3x - 1$ Find $f^{-1}(x)$

1) Write function as $y \rightarrow y = 3x - 1$

2) Interchange x and $y \rightarrow x = 3y - 1$

3) Rearrange, make y subject $\rightarrow y = \frac{x+1}{3}$

$$\rightarrow f^{-1}(x) = \frac{x+1}{3}$$

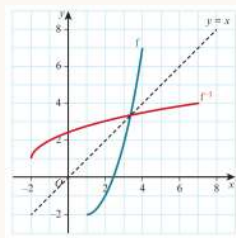


4. Graphs of functions and its inverse

= reflections of each other over the line $y = x$

* if f is self-inverse, $f(x) = f^{-1}(x)$

(e.g.: $f(x) = \frac{1}{x} = f^{-1}(x)$)

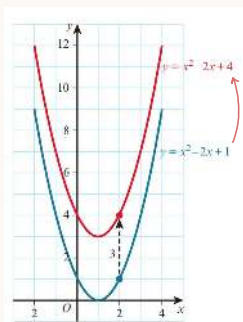


graph of f is symmetrical about the line $y = x$

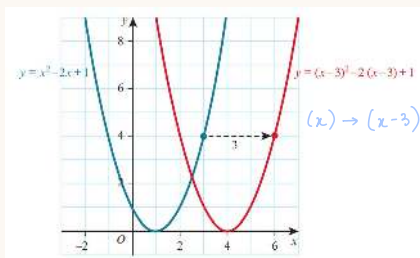
5. Transformations

$f(x-a) + b$	translation through $\begin{pmatrix} a \\ b \end{pmatrix}$
$af(x)$	stretch // y-axis, scale factor a
$f(ax)$	stretch // x-axis, scale factor $\frac{1}{a}$
$f(-x)$	reflection in the y-axis
$-f(x)$	reflection in the x-axis

1) Translations

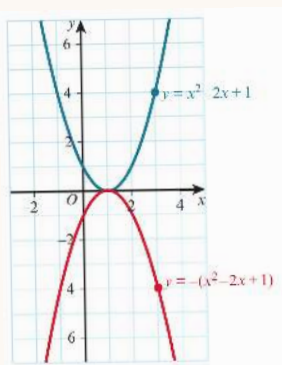


- vertical;
- $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$

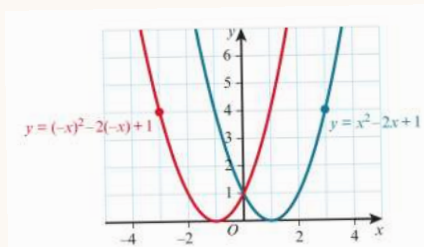


- horizontal ; $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$

2. Reflection

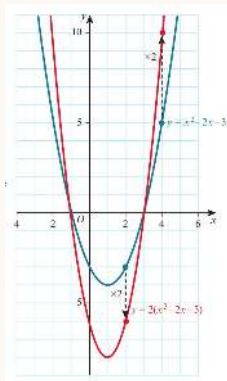


- over x-axis
- $-f(x)$

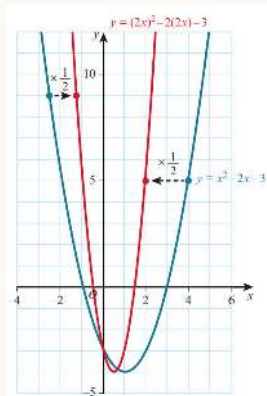


- over y-axis
- $f(-x)$

3. Stretch



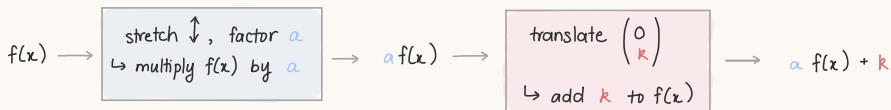
- $a f(x)$
- $2 f(x)$
- $\parallel y\text{-axis}$



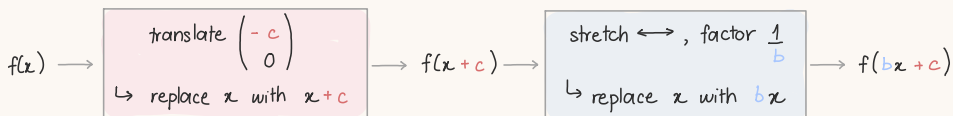
- $f(ax)$
- $f(2x)$
- $\parallel x\text{-axis}$

* combining transformations

• 2 vertical transformations



• 2 horizontal transformations



* c switches $(+/-)$

* b switches $(x/\div)$